

NEWTON 8 W, GA, USA

WMO: 740050

Lat: 31.3127N

Lon: 84.4706W

Elev: 54

StdP: 100.68

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 63828

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.3	(c) -2.5	(d) -10.5	(e) 1.5	(f) 2.8	(g) -7.9	(h) 1.9	(i) 3.1	(j) 3.7	(k) 19.4	(l) 3.1	(m) 16.2	(n) 0.2	(o) N/A	(p) 0.300

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 34.9	(d) 23.7	(e) 33.9	(f) 23.8	(g) 32.9	(h) 23.8	(i) 26.1	(j) 31.3	(k) 25.6	(l) 30.8	(m) 25.2	(n) 30.3	(o) 1.5	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 24.8	(b) 20.0	(c) 28.0	(d) 24.3	(e) 19.3	(f) 27.6	(g) 23.9	(h) 18.9	(i) 27.3	(j) 80.9	(k) 31.4	(l) 78.9	(m) 31.0	(n) 77.1	(o) 30.3	(p) 27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.0	2.7	2.2	-7.4	36.0	1.6	1.5	-8.6	37.1	-9.6	38.0	-10.5	38.9	-11.7	40.0
			-7.8	27.0	2.0	0.4	-9.2	27.2	-10.4	27.5	-11.5	27.7	-12.9	28.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	9.4	11.4	15.0	18.4	22.8	26.2	27.0	26.8	24.7	19.5	13.7	11.1
	DBStd	7.40	5.62	4.99	4.56	3.51	2.96	1.72	1.38	1.49	2.36	4.11	4.65	5.22
	HDD10.0	203	83	41	12	1	0	0	0	0	0	1	16	51
	HDD18.3	1068	282	199	121	41	6	0	0	0	1	37	151	231
	CDD10.0	3438	63	79	166	254	397	485	526	522	440	295	128	84
	CDD18.3	1262	3	4	17	44	144	235	268	263	191	73	13	6
	CDH23.3	11480	9	33	172	484	1410	2227	2491	2320	1590	622	98	23
CDH26.7	4759	0	2	23	106	552	1038	1157	1037	661	173	11	0	
Wind	WSAvg	0.8	0.9	1.0	1.1	0.9	0.9	0.8	0.8	0.7	0.7	0.7	0.7	0.8
Precipitation	PrecAvg	1259	121	101	143	78	58	137	152	121	119	72	78	89
	PrecMax	1753	379	171	284	187	155	250	250	237	367	218	229	164
	PrecMin	830	24	15	9	16	3	38	92	56	21	9	11	22
	PrecStd	239	77	47	90	44	39	65	51	51	86	49	53	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	26.5	28.9	30.8	34.3	36.3	35.9	35.7	35.0	32.1	28.0	25.5
		MCWB	18.6	19.3	18.1	20.6	21.3	22.5	24.8	23.7	23.0	22.2	20.1	19.8
	2%	DB	22.8	24.2	27.1	29.4	32.6	34.9	34.4	34.2	33.1	30.2	26.1	23.8
		MCWB	17.5	17.7	17.9	19.6	21.3	23.6	24.5	24.4	22.9	20.2	18.9	18.7
	5%	DB	20.7	22.0	25.3	27.9	31.2	33.5	33.4	33.0	31.8	28.7	24.1	21.8
		MCWB	17.1	17.4	17.5	18.8	21.0	23.8	24.5	24.5	22.9	19.6	18.1	18.8
	10%	DB	18.8	20.1	23.6	26.4	30.0	32.1	32.3	31.9	30.7	27.0	22.3	20.0
		MCWB	15.8	16.1	16.6	18.1	20.8	23.7	24.5	24.5	22.8	19.3	17.3	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	20.6	20.7	22.6	24.1	26.3	26.5	26.8	25.7	24.6	22.9	21.8
		MCDB	22.6	24.2	24.5	27.1	28.4	32.2	31.8	31.9	30.5	28.3	25.2	24.3
	2%	WB	19.0	19.7	19.9	21.6	23.4	25.5	25.9	26.2	25.0	23.5	21.5	20.6
		MCDB	21.3	23.1	23.7	26.5	28.2	31.3	31.3	31.1	29.7	26.9	23.7	22.5
	5%	WB	17.9	18.7	19.0	20.8	22.8	25.0	25.6	25.7	24.5	22.5	20.2	19.6
		MCDB	20.2	22.1	23.2	25.1	27.9	30.6	30.9	30.6	29.4	26.1	22.7	21.5
	10%	WB	16.5	17.3	18.0	20.0	22.2	24.5	25.2	25.2	23.9	21.4	18.4	18.2
		MCDB	18.9	20.3	22.4	24.3	27.4	29.7	30.4	30.0	28.7	25.2	21.9	20.3
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.0	13.5	13.8	13.1	11.3	10.7	10.2	11.3	13.3	14.3	12.5
		MCDBR	13.1	14.3	14.9	15.1	14.3	13.8	12.2	11.9	13.0	14.5	14.7	12.7
		MCWBR	8.1	8.1	6.9	6.4	5.0	4.3	3.9	4.2	4.9	6.7	8.2	8.2
	5% WB	MCDBR	10.5	11.6	12.0	12.0	11.4	11.0	10.7	10.4	11.0	10.9	10.6	10.1
		MCWBR	7.6	7.8	6.4	6.3	4.6	4.4	4.2	4.2	4.5	5.6	6.6	7.5
Clear-Sky Solar Irradiance	taub		0.326	0.333	0.355	0.384	0.433	0.479	0.508	0.497	0.425	0.374	0.351	0.332
	taud		2.511	2.487	2.441	2.359	2.255	2.188	2.118	2.147	2.334	2.450	2.456	2.512
	Ebn at Noon		902	929	928	907	861	817	791	796	846	871	863	875
	Edh at Noon		87	98	110	124	139	148	158	152	121	101	92	83
All-Sky Solar Radiation	RadAvg		3.03	3.60	4.78	5.88	6.45	6.09	6.03	5.53	4.97	4.30	3.39	2.64
	RadStd		0.19	0.42	0.40	0.47	0.45	0.45	0.36	0.38	0.41	0.43	0.27	0.25

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.78	N/A	N/A	N/A	N/A	N/A	N/A	-172	+231	N/A
(47) Regional (38 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+102

Nomenclature: See separate page